Big Data

Q1.

Big Data means very large datasets. It usually has high volume and speed, and different types of data. It is characterized by high volume, velocity and variety of data, and requires advanced technologies to extract meaningful insights.

Q2.

MapReduce is a programming model used in Hadoop to process large datasets in a distributed manner across multiple computers. It works in two main steps: Map, which processes and converts input data into key-value pairs, and Reduce, which combines and summarizes the results to produce the final output.

Q3.

The 5 V's of Big Data are Volume (large amount of data), Velocity (speed of data generation), Variety (different data types), Veracity (data accuracy), and Value (usefulness of data)

Q4.

Big Data is mainly of three types: Structured data (data stored in tables like databases), Unstructured data (data without fixed format like images, videos, emails, and social media posts). Semi-structured data (data with some structure like XML, JSON).

Q5.

Hadoop is an open-source framework used to store and process large amounts of data across multiple computers. It uses HDFS for distributed storage and MapReduce for parallel data processing, making it reliable, scalable, and cost-effective.

Q6.

Hadoop (Hadoop Distributed File System) is a distributed file system designed to store and manage very large files across multiple machines in a Hadoop cluster. It provides high fault tolerance, scalability, and high throughput access to data.

Main components of HDFS Architecture:

i) NameNode
ii) DataNode
iii) Secondary NameNode

i) NameNode (Master Node)

- The namenode is the brain of HDFS.
- it stores metadata, not actual data.
- Metadata includes:
  - → File names
  - → File permissions
  - → Block locations
  - → Directory structure
- It decides where data blocks are stored in the cluster.

ii) DataNode (Slave Node)

- DataNodes store the actual data blocks.
- Each file is split into fixed-size blocks. (default 128 MB)
- DataNode perform:
  - Read and write operations
  - Block creation, deletion, and replication
- They regularly send heartbeat signals to the NameNode to show they are active.

iii) Secondary Name Node :-

- It is not backup NameNode.
- It periodically merges:
  - FSImage
  - Editlogs

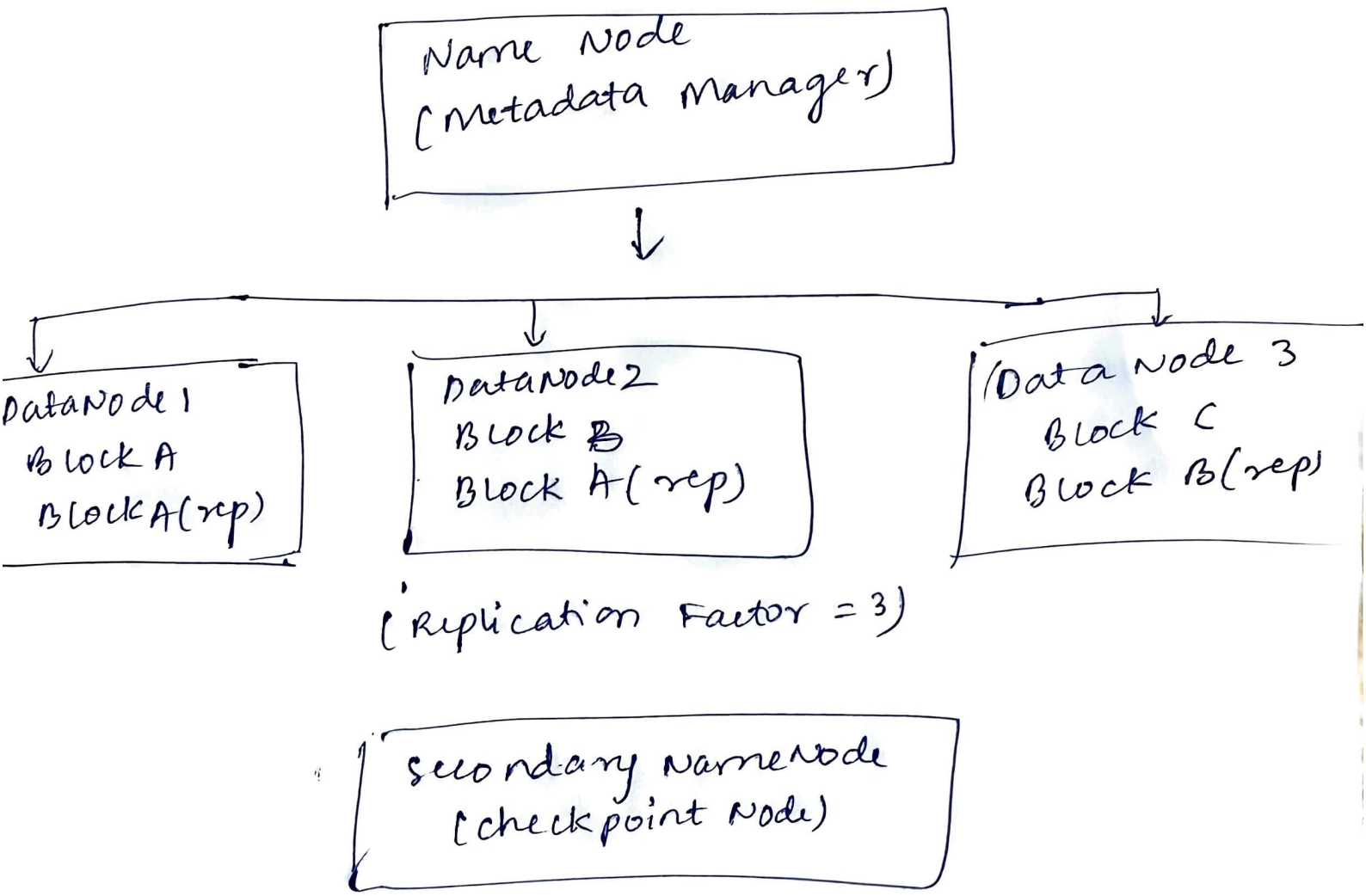
- This reduces the load on the NameNode and helps in faster recovery

Working of HDFS Architecture :-

1. File Write Process :-
   - Client requests the NameNode to store a file.
   - NameNode splits the file into blocks and selects DataNodes
   - Blocks are written to DataNodes
   - Data is replicated for fault tolerance.
2. File Read Process :-
   - Client requests the NameNode for block locations
   - NameNode returns the list of DataNodes.
   - Client reads data directly from the nearest dataNode.

Advantages :-

- Highly scalable
- Fault tolerant
- cost effective
- Suitable for large Data processing.


Name Node
(Metadata Manager)
DataNode 1
Block A
Block A(rep)
DataNode 2
Block B
Block A(rep)
DataNode 3
Block C
Block B(rep)
(Replication Factor = 3)
Secondary NameNode
(checkpoint Node)

Q7.

Replication is a core feature of HDFS that helps in storing multiple copies of the same data block on different DataNodes. It plays a major role in providing fault tolerance and high data availability in a Hadoop cluster.

Why Replication is Important in HDFS:

1. Fault Tolerance :-
   - Hardware failure are common in large clusters.
   - Replication ensures data is not lost when a DataNode fails.

2. High Data Availability :-
   - Multiple copies allow data to be accessed even if one copy becomes unavailable.

3. Reliable Data storage :-
   - Data remains safe even during node crashes or network failures.

4. Improved Read performance :-
   - Clients can read data from the nearest replica, reducing latency.

5. Supports Large Scale Systems:.

- Helps Hadoop scale across thousands of low-cost machines

How HDFS Ensures Data Availability when a DataNode fails :-

1. Replication Factor:-

- Each data block is stored on multiple DataNodes.
- Default replication factor is 3.
- Replicas are stored on different nodes and racks.

2. Heartbeat Mechanism:-

- DataNodes send heartbeat signals to the NameNode at regular intervals.
- If a DataNode stops sending heartbeats, it is marked as dead.

3. Automatic Replica Creation:-

- When a DataNode fails:
- The NameNode detects missing replicas
- It instructs other datanodes to create new replicas.

4. Rack Awareness :-

- Replicas are distributed across different racks
- Ensures data Availability even if an entire rack fails.

5. Client Read Redirection :-

- If one DataNode fails during a read operation
- The client automatically switches to another DataNode holding replica.
- No manual intervention is required.